

Jacopo Minniti

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EDUCATION

Minerva University

Sep 2023 – May 2027

Bachelor of Science in Computational Sciences

- Concentrations: AI & Mathematics; Minor: Neuroscience. **GPA 4.00 (major); 3.94 (overall).**
- Relevant Courses: ML, Bayesian Stats, Real Analysis, ODEs, Optimization, Distributed Systems.

PUBLICATIONS & PREPRINTS

Neural Dynamics as the Composition of Quantized Units

Submitted to ICLR 2027

- Modeled neural learning as the ordered acquisition of discrete, reusable computations, predicting their acquisition order and linking their composition to scaling laws.
- On the number naming dataset, recovered an interpretable model from a Transformer's training trajectory using only 35.7% of its parameters; used a specular algorithm to raise exact-sequence accuracy from 50.5% to 89.9%.

Learned Relay Representations for Forward-Thinking Discrete Diffusion Models

SPiGM Workshop at ICML 2026; Submitted to ICLR 2027

- Introduced learned relay representations trained via Backpropagation Through Time to pass continuous latent states across denoising steps, enabling forward-thinking in discrete diffusion models.
- Demonstrated significant reasoning and coding gains, improving LLaDA-8B accuracy on MATH500 by 7.6% and HumanEval accuracy by 8.54%.

Challenges in Inference Time Scaling with Uncertainty Aware Tree Search

ICLR 2026, Workshop on Scaling Post-training for LLMs

- Designed inference algorithm to reallocate CoT budget to ambiguous states using a process uncertainty model.
- Showed that entropy is a reliable uncertainty metric in-distribution but fails under search-induced distribution shift due to overoptimization, with correlation to rollout variance collapsing from $r = 0.62$ to 0.03.

Effects of Semantic Relatedness and Vocabulary Knowledge on New Word Meanings

CLDC 12; Submitted to Journal of Memory and Language

- Conducted experiments with 300+ participants to evaluate the acquisition of 32 novel meanings for familiar words.
- Demonstrated that semantic relatedness increased recall accuracy ($p < .001$) and that higher baseline vocabulary scores predicted improved form-meaning association ($R = 0.2$), mitigating age-related learning declines.

PROFESSIONAL EXPERIENCE

Member of Technical Staff, [Sakana AI](#), Tokyo

May 2026 – Sep 2026

- Developed a theory and experimental framework for decomposing transformer learning into discrete, reusable computations and ordered causal dependencies.
- Used this framework to guide training of a transformer, improving exact-sequence accuracy by 75.3% on NumberNaming dataset.

Research Assistant, [University of Toronto](#), [Tim G. J. Rudner Lab](#), Remote

Sep 2025 – May 2026

- Introduced stateful information flow across diffusion denoising steps, improving long-horizon planning and reducing inference latency by up to 32%.

AI Safety Researcher, [University of Chicago](#), [XLab](#), Chicago

Jun 2025 – Aug 2025

- Probed LLM latent states for reasoning uncertainty, predicting step-level CoT entropy with 89% AUROC on Qwen3.

SKILLS & ACCOMPLISHMENTS

- *Programming Languages:* Python, Java, JavaScript, Dart, Matlab, SQL
- *Technical Skills:* AWS, Linux, scikit-learn, PyTorch, React, Docker, Git, RAG, EEGLAB
- *Awards:* SAT (1580/1600), [Winner at NASA Space Apps Mountain View](#), [Double Prize at CalHacks 2023](#)
- [Talk at the University of Chicago](#) on prospects of using machine learning to categorize inductive biases.